Project A: Content-Grounded GenAI Engine (POC)

1) Background and purpose

McGraw Hill is exploring AI-enabled tooling that helps instructors (and potentially learners) generate and adapt learning materials while staying grounded in adopted content and operating within appropriate guardrails. This Proof of Concept (POC) will deliver a **reusable, content-grounded GenAI "engine"** that powers instructor workflows and provides a foundation for future productization.

This POC intentionally **avoids deep platform integration** (e.g., Connect/SimNet) and instead focuses on **demonstrating credible end-to-end capability** suitable for internal stakeholder demos and future investment decisions.

2) Objectives (POC outcomes)

By the end of the performance period, Project A will deliver:

1. A working **content-grounded GenAI engine** that can:
    - ingest OER instructional content,
    - retrieve relevant sources (RAG),
    - generate responses with **citations/provenance**, and
    - enforce **scope/guardrail behaviors** appropriate for a POC.
2. A **stable API and configuration model** that Project B can consume to implement instructor tools without re-building the backend.
3. An **engine validation suite** (quality + safety evaluation) and an **end-of-POC report** that documents feasibility, constraints, risks, and next steps for productization.

3) In-scope scope (what will be built)

A. Content and ingestion (OER-first)

- Select **one OER textbook / OER content set** (with sponsor concurrence) to serve as the POC content foundation.
- Implement ingestion pipeline:
    - content import (PDF/HTML/text),
    - chunking strategy,
    - embedding generation,
    - indexing in a vector store,
    - metadata tagging (chapter/section) to enable traceable citations.

B. Retrieval-Augmented Generation (RAG) service

- Implement a retrieval layer that supports:
    - top-k retrieval by semantic similarity,
    - source attribution (chapter/section citations),
    - configurable retrieval settings per use case (e.g., conservative vs broader retrieval).
- Implement answer generation that:
    - uses retrieved chunks,

    - returns citations,
    - supports “course-only mode.”

C. “Template-driven” GPT configuration (ring-fenced)

- Provide a **template-based configuration model** (not freeform prompt engineering) enabling **2–3 approved templates** for the POC, such as:
    1. Title/Chapter Q&A (course-only tutor)
    2. Role/Scenario Coach (course-only, instructor-controlled)
    3. Instructor task assistant (e.g., “draft lesson outline aligned to objectives”)
- Templates are “guardrailed recipes”: fixed structure + limited instructor parameters (audience, tone, learning objectives, allowed resources, external toggle).

D. Publish/access controls (POC-level)

- Provide a minimal mechanism to:
    - “publish” a configured experience,
    - restrict access to a limited set of users (e.g., invite link or simple allowlist),
    - log usage events (no PII required; see constraints).

Note: This is **not** full LMS roster sync or SSO—POC-level access control only.

E. Guardrails (POC-appropriate “proof,” not production)

- Implement baseline controls consistent with “proof of guardrails”:
    - explicit out-of-scope refusal behaviors,
    - “course-only” grounding requirement,
    - basic prompt-injection resilience patterns (e.g., ignore instructions embedded in retrieved content),
    - output handling safeguards (e.g., prevent returning system prompts/keys; avoid unsafe content categories).
- Provide documentation describing guardrails and known limitations.

F. API contract and developer usability

- Deliver a versioned **Engine API contract** that supports Project B workflows:
    - “generate lesson outline” request shape,
    - “transform assessment” request shape,
    - “retrieve sources” support,
    - configuration endpoints (template selection, parameters),
    - telemetry endpoints (usage counts, latency logs).
- Provide API documentation and a minimal SDK/client example for Project B.

G. Engine validation plan and evaluation artifacts

- Develop a lightweight evaluation suite including:
    - a **golden set** of representative prompts aligned to the selected OER content,

    - citation correctness checks (answer supported by cited sources),
    - retrieval relevance checks,
    - out-of-scope refusal tests,
    - prompt-injection/adversarial prompt tests,
    - basic performance checks (latency, error rates).
- Produce evaluation snapshots tied to sponsor presentation milestones.

4) Out of scope (explicit exclusions)

- Production deployment in McGraw Hill systems (Connect/SimNet/ALEKS integration).
- Full enterprise security/compliance certification.
- Full SSO/roster synchronization.
- Fine-tuning custom models as a core deliverable.
- Per-GPT dedicated VM/container "spin-up" architecture.
- Building a full agent platform; "agents" are limited to workflow orchestration steps within the engine.

5) Assumptions and constraints

- **Content constraint:** POC uses **OER** content unless sponsor later provides approved alternate content and permissions.
- **Integration constraint:** No platform integration beyond a standalone demo environment.
- **Accessibility:** No WCAG compliance requirement.
- **Data handling:** POC does not require storage of student PII; usage logs should be anonymized/aggregated where possible.
- **Model/provider provisioning:**
    - Early development may use **free-tier experimentation** where permitted and appropriate.
    - POC **must not rely on free-tier availability** for formal sponsor demos or instructor pilot usage.
    - By the first formal sponsor presentation (week of **March 23**), the team will move to a **stable, project-owned or sponsor-provided paid/quota-controlled** environment with documented usage limits.
    - No personal student accounts for shared environments/keys.

6) Success metrics and acceptance criteria (POC "Definition of Done")

Project A will be accepted when:

- **Engine functionality:** Ingests selected OER content and answers queries grounded in retrieved sources with citations.
- **Template operation:** Supports 2–3 approved templates with constrained parameters.
- **Guardrails:** Demonstrates out-of-scope refusals and "course-only" grounding in agreed test cases.

- **API stability:** Provides an Engine v0 API consumed by Project B (integration demonstrated).
- **Validation suite:** Delivers evaluation artifacts and results for agreed test set at each formal sponsor presentation.
- **Operational readiness (POC):** Runs reliably for demo usage within defined quotas and logs basic telemetry.

7) Quality & evaluation plan

Project A's evaluation approach is **engine validation** rather than end-user usability testing:

- **Functional validation:** regression checks on golden prompt set
- **Grounding validation:** citation coverage + faithfulness sampling checks
- **Safety validation:** refusal behavior + injection tests
- **Integration validation:** API contract tests + end-to-end calls from Project B
- **Performance validation:** latency/error rate monitoring during demos/pilot window

Outputs: short evaluation report per formal presentation + final consolidated evaluation report.

8) Governance, roles, and cadence

- **Sponsor (McGraw Hill):** Product Owner; provides feedback, approves milestones, recruits instructors for any pilot where applicable.
- **Faculty Advisor (Mohammed Shafae):** Technical/program oversight; approves scope changes and academic alignment.
- **PM (Anirudha Soni):** Schedule/risk/communications; runs backlog and change log; ensures A→B dependency management.
- **Student team:** design/build/test/document.

Cadence:

- **Bi-weekly sponsor meetings** (demo + decision + risk review).
- Internal weekly team standups as needed.
- Scope changes recorded in a simple **change log**; sponsor approval required for scope additions.

9) Schedule and milestones

Sponsor meetings occur bi-weekly; formal presentations are fixed at the weeks specified.

Phase 0 — SOW Revision (Pre-execution)

- **1/26 – 2/13:** Revised SOW and milestone plan finalized and approved (deliverable due **2/13**).

Phase 1 — Execution (Students begin week of 2/16; performance period ends 8/12)

- **Sprint 1:** Architecture + OER selection + API contract draft + mock endpoints for Project B
- **Sprint 2:** Ingestion pipeline v0 + baseline RAG + citation formatting
- **Sprint 3 (Presentation week of 3/23): Presentation #1** — Engine v0 demo, API live, baseline guardrails, initial evaluation snapshot; stable provider path established
- **Sprints 4–6 (to Presentation week of 5/4):** improved retrieval quality, template refinement, telemetry, expanded evaluation suite
- **Sprint 6 (Presentation week of 5/4): Presentation #2** — Engine v1, stronger grounding/guardrails, integration readiness status
- **Sprints 7–9 (to Presentation week of 6/15):** hardening, injection testing, evaluation expansion, integration support for Project B pilot
- **Sprint 9 (Presentation week of 6/15): Presentation #3** — Engine v2, pilot support readiness, mid-pilot monitoring
- **Sprints 10–13 (to Presentation week of 8/10; end 8/12):** stabilization, documentation, final evaluation report, roadmap
- **Sprint 13 (Presentation week of 8/10): Presentation #4** — Final demo + final report; **8/12** handoff complete

10) Project A deliverables (summary)

- Revised SOW (Phase 0) — due 2/13
- OER content selection + ingestion plan
- Engine v0 API contract + mock server (early execution)
- Ingestion pipeline + vector index
- RAG service with citations
- Template library (2–3 templates)
- Guardrails baseline + documentation
- Telemetry/usage logging (POC-level)
- Validation suite + milestone reports
- Final architecture + handoff package + roadmap

11) Dependency handoff to Project B

Project A must deliver:

- **API contract + mock responses early** (to unblock Project B)
- **Engine v0 live** by Presentation #1 window (week of 3/23)

- **Stable provider/quota path** by Presentation #1 (Pattern B requirement)
- Ongoing compatibility with Project B through a versioned change log